

Silicon Solar Cells: Principles and Performance

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1. History

The first practical silicon solar cell was demonstrated in 1954 at **Bell Labs** by **Daryl Chapin**, **Calvin Fuller** and **Gerald Pearson**; it converted about 6% of sunlight into electricity. Early cells powered satellites such as *Vanguard 1* (1958). In 1961 **William Shockley** and **Hans-Joachim Queisser** derived the detailed-balance limit for a single p-n junction, which is about 33.7% for a band gap of 1.34 eV under standard sunlight.

2. Device equations

2.1 Current-voltage characteristic

An illuminated solar cell is modelled as a current source in parallel with a diode:

$$I = I_L - I_0 \left[\exp \left(\frac{qV}{nkT} \right) - 1 \right]$$

Setting $I = 0$ gives the open-circuit voltage:

$$V_{oc} = \frac{nkT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right)$$

2.2 Efficiency

The fill factor FF and the power conversion efficiency η are defined as:

$$FF = \frac{P_{max}}{V_{oc} I_{sc}}, \quad \eta = \frac{FF \cdot V_{oc} \cdot I_{sc}}{P_{in}}$$

where:

- I_L — photogenerated current, approximately equal to the short-circuit current I_{sc} ;
- I_0 — dark saturation current of the diode;
- n — ideality factor; k — Boltzmann constant; T — cell temperature;
- P_{in} — incident light power, 1000 W/m^2 under standard test conditions (STC).

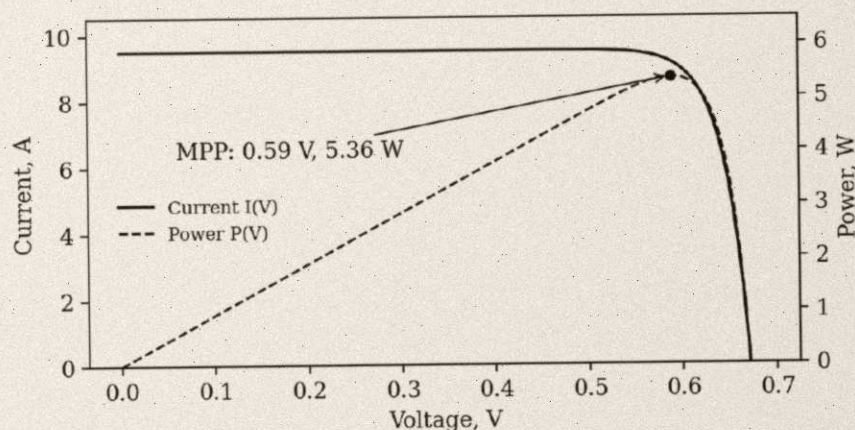


Figure 1. Calculated I - V and P - V curves of a 156 mm silicon cell ($I_L = 9.5 \text{ A}$, $I_0 = 5 \cdot 10^{-11} \text{ A}$, $n = 1$, $T = 300 \text{ K}$).

3. Photovoltaic technologies

Table 1. Comparison of photovoltaic technologies (approximate values)¹

Group	Technology	Absorber		Typical module efficiency, %	Market status
		Band gap, eV	Band gap type		
Crystalline silicon	Monocrystalline Si	1.12	indirect	20 – 23	dominant
	Multicrystalline Si	1.12	indirect	16 – 18	declining
Thin film	CdTe	1.45	direct	18 – 19	utility scale
	CIGS	1.0 – 1.7	direct	14 – 17	niche
	Amorphous Si	1.7	amorphous	6 – 9	legacy
Emerging	Perovskite	1.5 – 1.6	direct	—	early commercial

4. Worked example

A monocrystalline cell of 156×156 mm under STC has:

- 1 $V_{oc} = 0.67$ V and $I_{sc} = 9.5$ A;
- 2 fill factor $FF = 0.80$, so $P_{max} = 0.80 \times 0.67 \times 9.5 \approx 5.1$ W;
- 3 incident power $P_{in} = 1000 \text{ W/m}^2 \times 0.0243 \text{ m}^2 \approx 24.3$ W;
- 4 efficiency $\eta \approx 5.1 / 24.3 \approx 21\%$.

5. Conclusions

- Crystalline silicon dominates the market thanks to mature manufacturing and low cost.
- Efficiency is limited by the band gap: the Shockley–Queisser limit is about 33.7%.
- Tandem cells, e.g. perovskite on silicon, can exceed the single-junction limit.

¹ Efficiency figures are typical values for commercial products and vary between manufacturers.